



AI-Based Exam Proctoring and Evaluation Platform

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Introduction

Addressing exam integrity challenges with AI-powered proctoring



Rapid growth of online education raises concerns about maintaining **exam integrity**



Traditional proctoring is **labor-intensive** and lacks scalability for large candidate pools



AI-based proctoring leverages computer vision techniques like **OpenCV** and AI models for automated monitoring



System detects suspicious activities such as **looking away, multiple faces, and excessive movement**



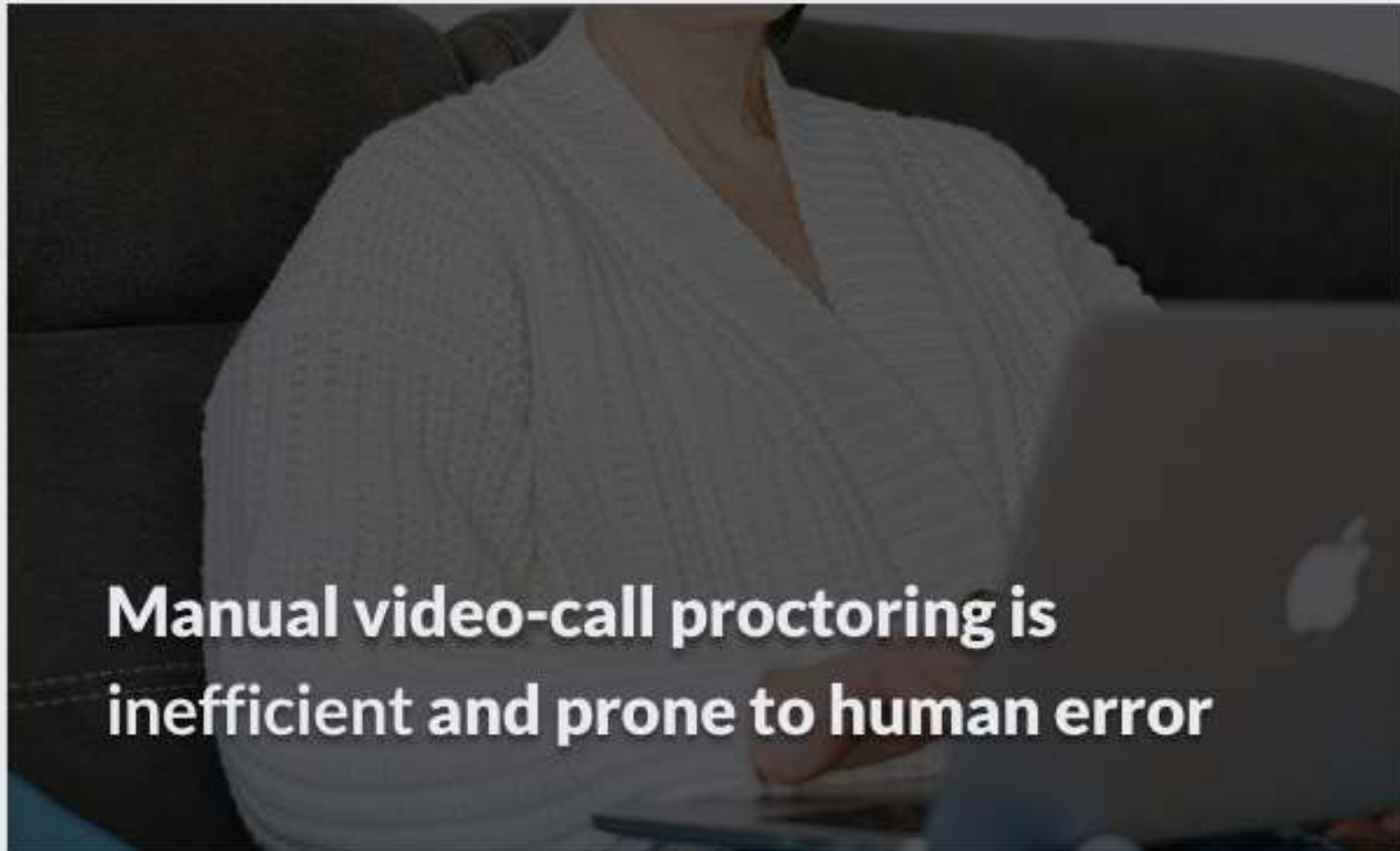
Enables **efficient, fair, and real-time** monitoring during online exams

Motivation

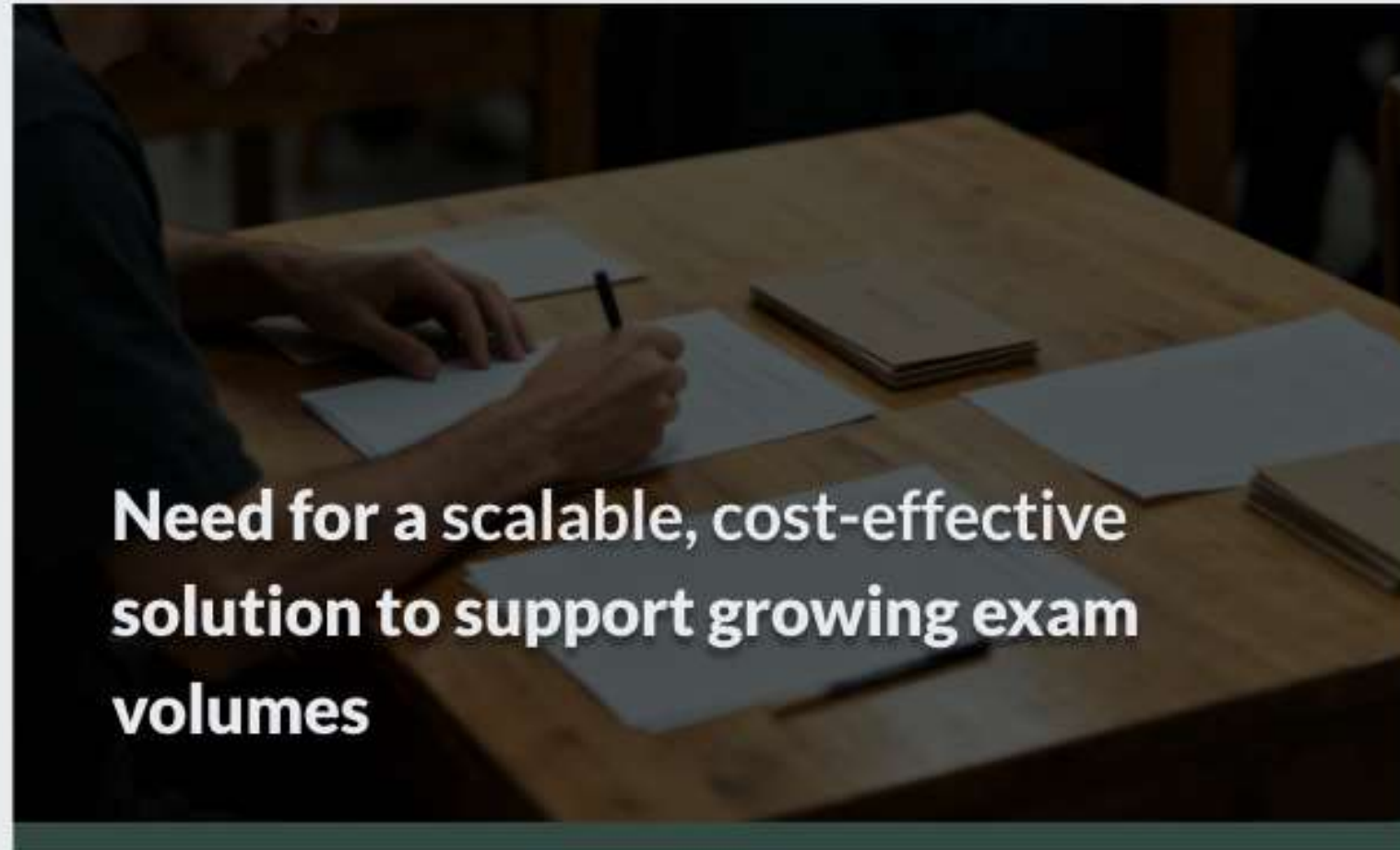
Driving Fairness and Efficiency in Online Exam Proctoring



Increasing online exam malpractice threatens fairness and academic integrity



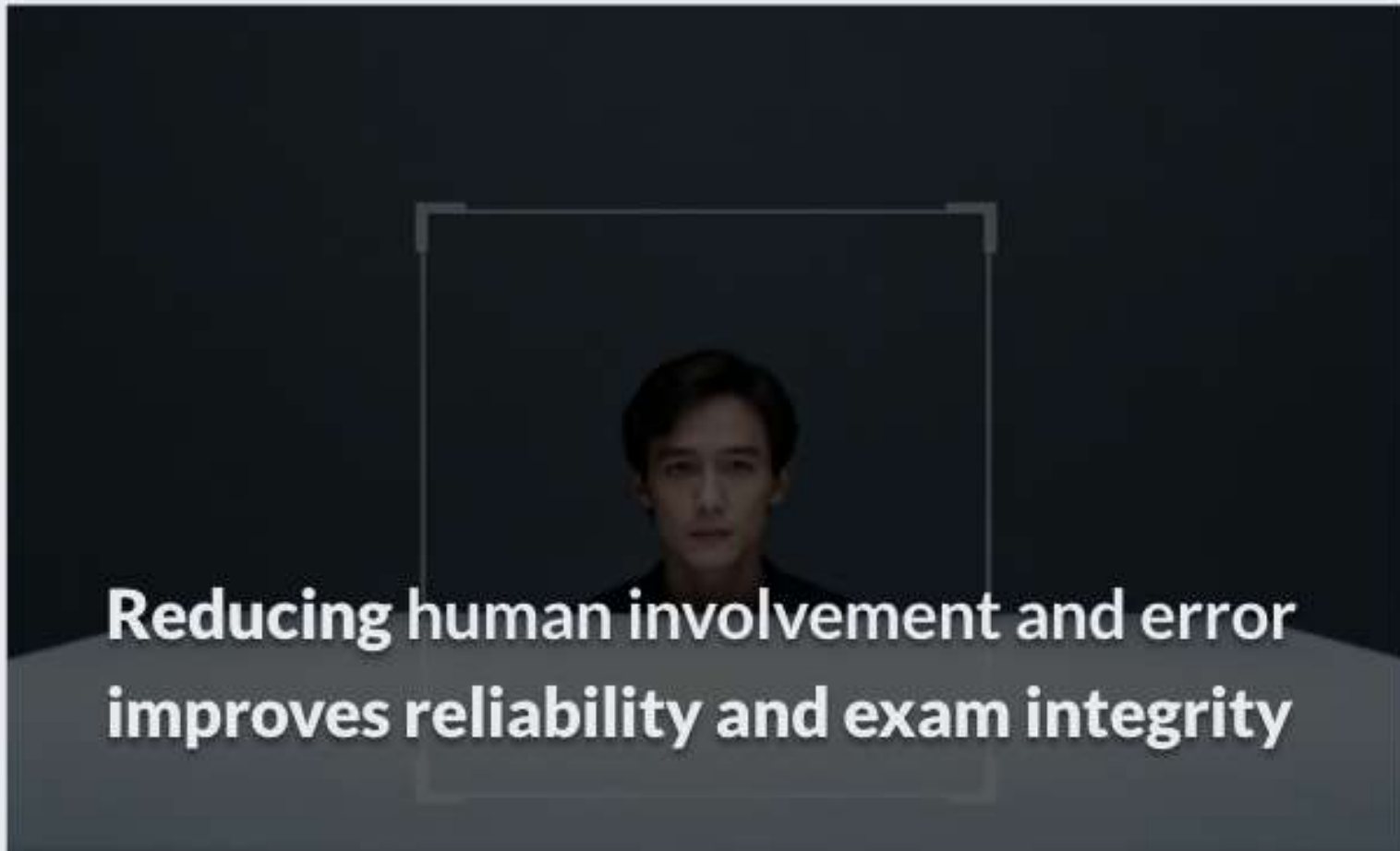
Manual video-call proctoring is inefficient and prone to human error



Need for a scalable, cost-effective solution to support growing exam volumes



AI-driven systems offer automated monitoring with higher accuracy and fairness



Reducing human involvement and error improves reliability and exam integrity



Our project applies AI and computer vision to responsibly improve online exam integrity

Literature Survey

Key AI Proctoring Studies (2019–2024) Shaping Real-Time Exam Systems



Paper Title	Year	Source
AI-Based Face Detection in Exam Proctoring	2019	Journal of Computer Vision
Gaze Tracking for Online Exam Integrity	2020	IEEE Transactions on Affective Computing
Emotion Recognition to Detect Cheating	2020	ACM Multimedia Conference
Deep Learning Approaches in Proctoring	2021	Neural Processing Letters
OpenCV-Based Real-Time Monitoring	2021	International Journal of AI
Multi-Person Detection in Exam Settings	2022	Pattern Recognition Letters
Hybrid Gaze and Facial Analysis Models	2023	AI in Education Journal
Lightweight Real-Time Proctoring Systems	2023	Sensors
Comprehensive Computer Vision in Exams	2024	Computer Vision and Image Understanding

State-of-Art Evaluation (Gap Analysis Table)



Comparing Algorithms, Features, and Limitations Across Proctoring Solutions

Work	Key Algorithms	Features	Limitations
Prior Solutions (General)	Face detection, gaze tracking, or motion detection separately	Single modality monitoring: face, gaze, or motion	Limited scope; lacks comprehensive detection; higher latency
Our System	Face, eye, gaze, and multi-person detection using lightweight OpenCV and Dlib	Real-time, comprehensive monitoring with multiple biometric cues	Lower latency and better resource efficiency

Problem Statement

Ensuring Exam Integrity Through Real-Time AI Monitoring



Develop an AI proctoring system that analyzes webcam video to detect **real-time suspicious behaviors** such as looking away or multiple faces



Automatically alert users and log warnings for **stakeholder review**



Ensure exam integrity through **automated monitoring** and detailed behavior tracking



Implement within a **lightweight, efficient framework** to minimize system impact

Objectives

Establish core goals to ensure robust and user-friendly monitoring



Implement real-time **face and eye tracking** using OpenCV



Detect and alert when candidates **look away** from the screen



Identify **multiple faces** simultaneously to trigger warnings



Maintain a detailed **warning log** for audit and review



Provide **live feedback** through frontend animations for usability



Ensure system **reliability, usability, and effective monitoring**

Social Impact (Applications)

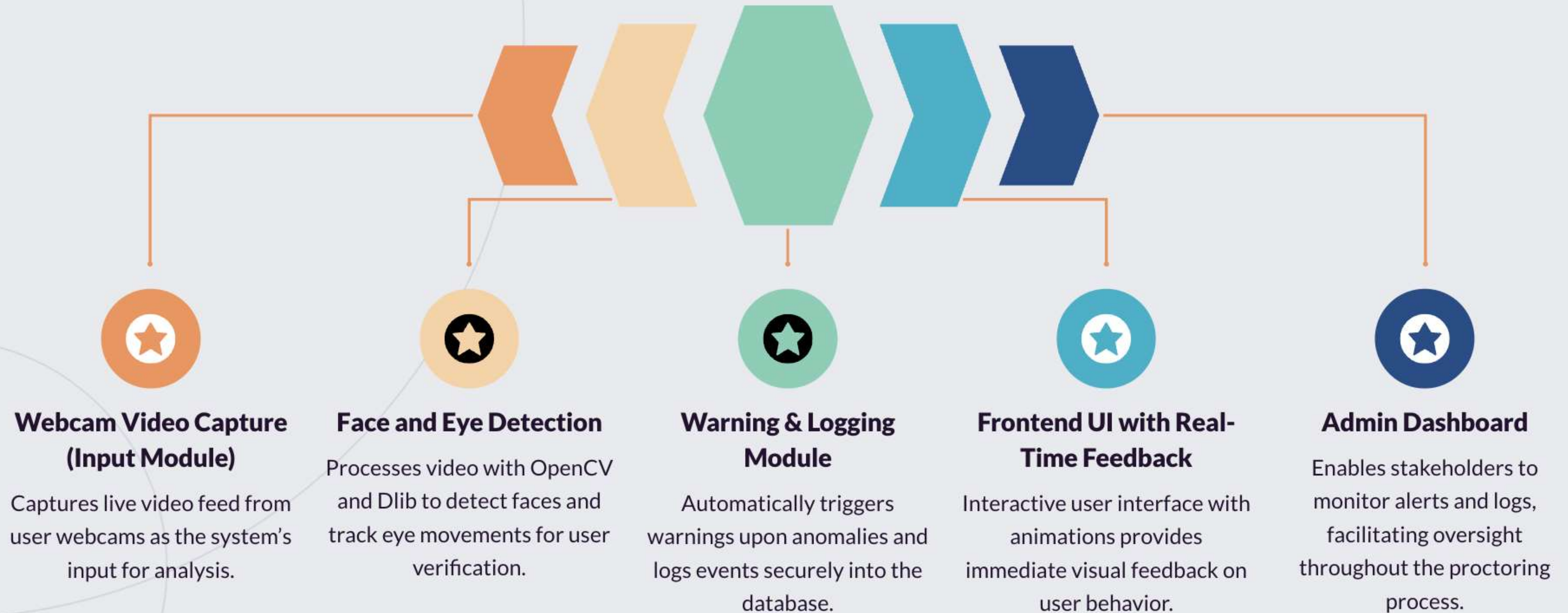
Enhancing fairness, engagement, and productivity across diverse domains



- 1 **Ensure fairness and prevent cheating in online examinations**
- 2 **Track candidate engagement during virtual interviews**
- 3 **Monitor student focus on e-learning platforms**
- 4 **Assess attentiveness in corporate training sessions**
- 5 **Detect inattentiveness or absence in remote work setups to encourage productivity and integrity**

System Architecture

Modular, Automated Workflow for Real-Time Proctoring



Feasibility and Scope

Key requirements, timeline phases, and risk mitigation strategies for project success



Requirements

Software: Python, OpenCV, Flask or Django backend, HTML/CSS/JS frontend

Hardware: 720p+ webcam, 4GB RAM, i3 or higher CPU

Timeline Phases

10-week span covering survey, implementation, integration, testing, documentation

Risks & Mitigation

Challenges: poor lighting and false positives

Mitigation: backup options including MediaPipe or cloud-based models

Proposed Algorithm

Stepwise process from data capture to real-time alert and logging



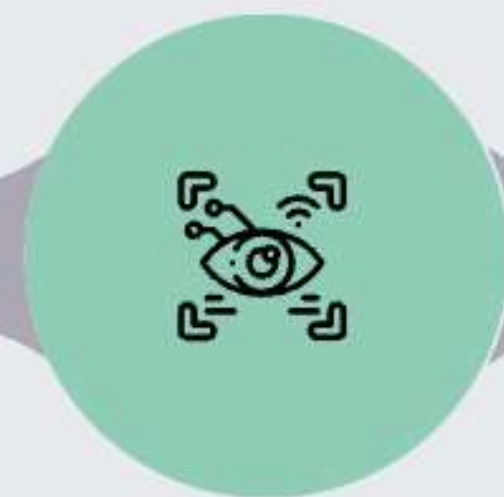
Capture Webcam Video Frames

Continuously acquire live video input from the webcam for analysis.



Detect Faces Using Haar Cascades

Apply Haar cascade classifiers to identify faces within each video frame.



Detect Eyes and Estimate Gaze Direction

Locate eyes and compute gaze vector to assess user's visual attention.



Trigger Warnings on Deviations

Generate alerts if gaze deviates beyond thresholds or multiple faces appear.



Log Warning Events

Record warnings for audit and review to track monitoring incidents.



Update User Interface with Live Animation

Reflect real-time detection status through live animations for user feedback.

Advantages and Limitations

Balancing real-time benefits with environmental and hardware constraints



Advantages

- Real-time monitoring of user actions
- Lightweight, fast execution using OpenCV
- User feedback animations enhance interaction
- No need for additional hardware setups
- Comprehensive logs available for review

Opportunities

- Optimize lighting detection algorithms
- Improve accuracy with diverse eyewear profiles
- Integrate with advanced cheating detection systems
- Expand user feedback customization
- Leverage logs for predictive analytics



Limitations

- Dependent on proper lighting conditions
- Accuracy drops with spectacles or low-quality webcams
- Cannot detect advanced cheating such as hidden screens

Threats

- Variability in webcam quality across users
- Evolving cheating techniques beyond detection scope
- User privacy concerns limiting feature adoption
- Lighting inconsistencies causing false readings

Test Cases (multiple faces & no face)

AI PROCTORING SYSTEM

TIME: 0:01:22

STUDENT: Mahesh

STATUS PANEL

- Faces: 2 (Multiple!)
- Multi Events: 5
- Gaze Dir: Center
- Eyes: Open
- Phone: Not Detected
- Phone Events: 2

GAZE COMPASS + CAMERA FEED

SUMMARY

Attention: 100.0%

Violations:

Total: 28

Recent Events:

23:16:55 ??? Browser Navigation Blocked (d

23:16:56 ??? Browser Navigation Blocked (d

23:16:56 ??? Multiple People (2 faces)

23:16:57 ??? Keyboard (3)

CAMERA: ??? AUDIO: ??? FPS: 19.1

[ESC] Exit [P] Preview

AI PROCTORING SYSTEM

TIME: 0:00:57

STUDENT: Mahesh

STATUS PANEL

- Faces: 1
- Multi Events: 1
- Gaze Dir: Center
- Eyes: Open
- Phone: Not Detected
- Phone Events: 1

GAZE COMPASS + CAMERA FEED

SUMMARY

Attention: 62.0%

Violations:

Total: 17

Recent Events:

23:16:30 ??? Browser Navigation Blocked (d

23:16:31 ??? Look Left (x:0.37 y:0.76)

23:16:31 ??? Browser Navigation Blocked (d

23:16:32 ??? Keyboard (3)

CAMERA: ??? AUDIO: ??? FPS: 21.2

[ESC] Exit [P] Preview

Conclusion and Future Work

Summarizing achievements and outlining planned enhancements for our AI proctoring system



Developed a real-time AI proctoring system using OpenCV with **gaze deviation detection**

Plan to integrate **emotion recognition** to improve candidate state monitoring

Incorporate **deep learning models** to enhance detection accuracy and robustness

Develop **cloud-based analytics dashboards** for comprehensive performance insights

Add **voice detection** to monitor background conversations for improved exam integrity



THANK YOU